Supplementary Material for the Article: Adaptive Scattering Transforms for Playing Technique Recognition

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This document provides further details regarding the datasets and detection results for our work, "Adaptive Scattering Transforms for Playing Technique Recognition" [1].

A. Existing Computational Research on Playing Techniques

Besides the summary of existing computational research on playing techniques in our paper [1], we provide a complete list of playing techniques analysed and the corresponding methodologies applied in Table I.

B. Additional Information on the Datasets

Table II displays the number of ground truth frames, total duration of the ground truth frames, and duration of co-articulations for each playing technique in the CBFdataset. As can be seen, the number of samples for each type of playing technique is highly imbalanced. Portamento has 1010 samples while the number of trills has only 197. Compared to the duration of the whole dataset, 157 minutes, the co-articulations amount to only 5 minutes. To have results comparable to the extended datasets discussed in our paper [1] and to obtain a confusion matrix between different CBF techniques, we discard the co-articulation samples and perform multiclass classification.

Fig. 1 shows the comparison between a glissando played in isolation and those performed as part of full-length pieces. The isolated glissando consists of consecutive notes of approximately equal duration changing in step-wise motion, while the ones performed in music context vary from each other in different ways. Player 1 lengthened the first note for the glissando at position 1, while emphasising the beginning of the glissando at position 2 by increasing the intensity. Player 2 performed the glissando together with flutter-tongue. These variations demonstrate the importance of collecting playing techniques in performance contexts.

C. Additional Result Analysis

Table III list the frame sizes and dimensionalities of the proposed and existing scattering features in the binary classification (see Section VI-A of our paper [1]) of the seven CBF playing techniques. The proposed adaptive scattering transforms include three frequency-adaptive scattering representations, i.e. AdaTS, AdaTRS, and AdaTS+AdaTRS, and two direction-adaptive scattering operators, i.e. dJTFS-max and dJTFS-avg. The existing scattering representations are the standard scattering and the joint time–frequency scattering (JTFS). As can be seen from the table, for each of the discussed playing techniques, the existing scattering representations exhibit much higher dimensionalities than that of the

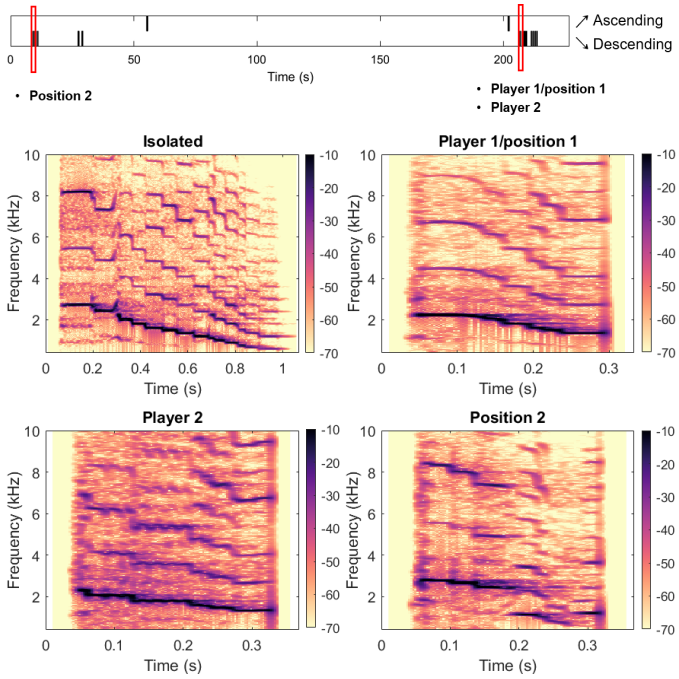

Fig. 1. Variations of glissandi between different players and different temporal positions in the performance of *Busy Delivery Harvest*.

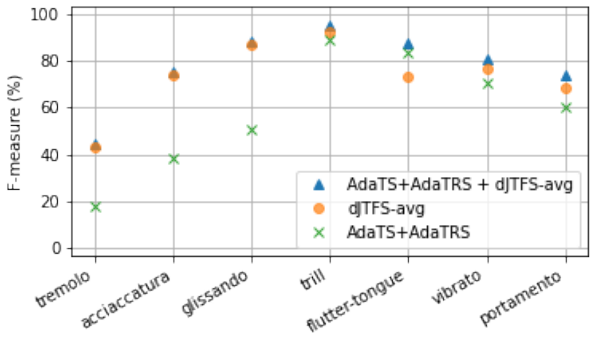

Fig. 2. Performance comparison of the recognition system on the CBFdataset using the proposed AdaTS+AdaTRS, the dJTFS-avg, and the concatenation of AdaTS+AdaTRS and dJTFS-avg as input.

TABLE I
SUMMARY OF EXISTING COMPUTATIONAL RESEARCH ON PLAYING TECHNIQUES

Instrument	Expressive patterns	Methodologies applied	Cite
Guitar	Ornamentation defined by rules	Dynamic time warping sequence matching	[2]
	Legato, glissando	Symbolic, aggregate approximation	[3]
	Muted, harmonic, bend, slide, hammer-on, pull-off	Rule-based detection	[4]
	Muting, vibrato, pull-off, hammer on, sliding, bending	Sparse cepstral and phase codes	[5], [6]
	Plucking: finger, picked, muted, slap; expression: harmonics, vibrato, bending, slide, etc.	Feature extraction, support vector machines with radial basis function kernel	[7], [8]
Piano	Trills	Independent component analysis	[9]
	Pdealling techniques	Supporter vector machines	[10]
Violin	Beginners' faults on playing techniques	K-nearest neighbours	[11]
	Flageolet, non-vibrato, pizzicato, sordino, spiccato, sul ponticello, sul tasto, tremolo	Sparse modelling	[12]
Drums	strike, buzz roll, drag, flam	Non-negative matrix factorization-based activation	[13]
	A variety of drum playing techniques	K-nearest neighbours, partial decision tree, etc.	[14]
Cello	Vibrato, tremolo, trill, glissando, etc.	Folded convolutional neural network	[15]
Irish flute	Cut, strike, crann, roll, shake	Rule-based method; hidden Markov models (HMMs)	[16]
Highland pipe	60 ornamentations	Dynamic time warping	[17]
Ney	Vibrato, kaydirma	Rule-based characterisation	[18]
Erhu	Vibrato, portamento	Filter diagonalisation method, HMMs	[19]
Guqin	None, vibrato, upward/downward/inverted-U/U protamento	Neural network	[20]
Chinese bamboo flute	Vibrato, tremolo, till, flutter-tongue, acciaccatura, portamento, glissando	HMMs, adaptive/joint time–frequency scattering	[21]–[23]
	Trill, flutter-tongue, tremolo	Synthesising	[24], [25]
Other	Vibrato, glissando	Rule-based methods	[26]
Multi-instruments	140 instrumental playing techniques (isolated)	Scattering transform	[27]
Vocal techniques	Vibrato, glissando, linear note change	COSFIRE filter	[28]
	17 different vocal techniques	VocalSet	[29]

TABLE II
NUMBER OF SAMPLES, DURATION OF GROUND TRUTH FRAMES, AND DURATION OF CO-ARTICULATIONS FOR THE SEVEN TECHNIQUES IN THE CBFDATASET. TOTAL DURATION OF THE CBFDATASET IS 157 MINUTES, INCLUDING 5-MINUTES CONTAINING CO-ARTICULATIONS

Techniques	Flutter-tongue	Trill	Vibrato	Tremolo	Acciaccatura	Portamento	Glissando
Number of samples	628	197	519	555	441	1010	490
Duration of ground truth (minute)	7.9	15.0	13.8	6.0	1.7	6.3	5.0
Duration of co-articulations (minute)	1.3	0.1	3.7	3.7	0	0.9	0.3

TABLE III
FRAME SIZES AND DIMENSIONALITIES OF THE PROPOSED AND EXISTING SCATTERING FEATURES IN THE BINARY CLASSIFICATION OF CBF PLAYING TECHNIQUES

PMTs	Frame size (ms)	Dimensionality			
		AdaTS	AdaTRS	AdaTS+AdaTRS	Standard
Flutter-tongue	46	42	42	84	222
Trill	186	133	70	203	1725
Vibrato	186	133	70	203	2074
Tremolo	186	133	70	203	2074

PETs	Frame size (ms)	Dimensionality		
		dJTFS-max	dJTFS-avg	JTFS
Acciaccatura	46	350	350	1490
Portamento	93	410	410	1486
Glissando	93	498	498	1770

proposed ones.

In Section VII-D of our paper [1], we detect all seven CBF playing techniques via a multiclass classifier using the concatenation of the proposed frequency-adaptive scattering (AdaTS+AdaTRS) and the direction-adaptive scattering (dJTFS-avg) representations as input. This is based on a performance comparison of the system using the AdaTS+AdaTRS only, the dJTFS-avg only, and the concatenation of these

two features. The concatenated feature achieves the highest macro averaged F-measure of 79.9%, as compared to that of 62.5% using only the AdaTS+AdaTRS, and 76.0% using only the dJTFS-avg. Fig. 3 shows the F-measures obtained in the multiclass classification of each playing technique using the three type of scattering representations. Although the dJTFS-avg obtains comparable result as that of the concatenated feature, the latter outperforms the former for all the seven playing techniques.

Fig. 3 and Fig. 4 show the confusion matrices obtained for recognising playing techniques in the SOL dataset and VocalSet. The SOL dataset presents few confusions between techniques. This may be attributed to the fact that the dataset consists of short clips of individual techniques. No technique is recorded on the same note(s) on the same instrument, which offers high intra-class variability for playing techniques. For the VocalSet, as can be seen from the normalised confusion matrix in Fig. 4 (b), the lip trill obtains the best score while the trill technique obtains the lowest. The trill is frequently confused with the vibrato. The straight technique is often misclassified as other techniques.

Fig. 5 and Fig. 6 give the confusion matrices obtained for playing technique recognition in the CBFdataset using mel-frequency cepstral coefficients (MFCCs) and the modulation power spectrum (MPS) as input features to the classifier, respectively. Both features achieve the best score for trill recognition. The MFCCs fail to detect tremolos while the MPS returns the lowest score for acciaccatura recognition. The confusion between vibrato and tremolo is also found in the recognition results for the MPS.

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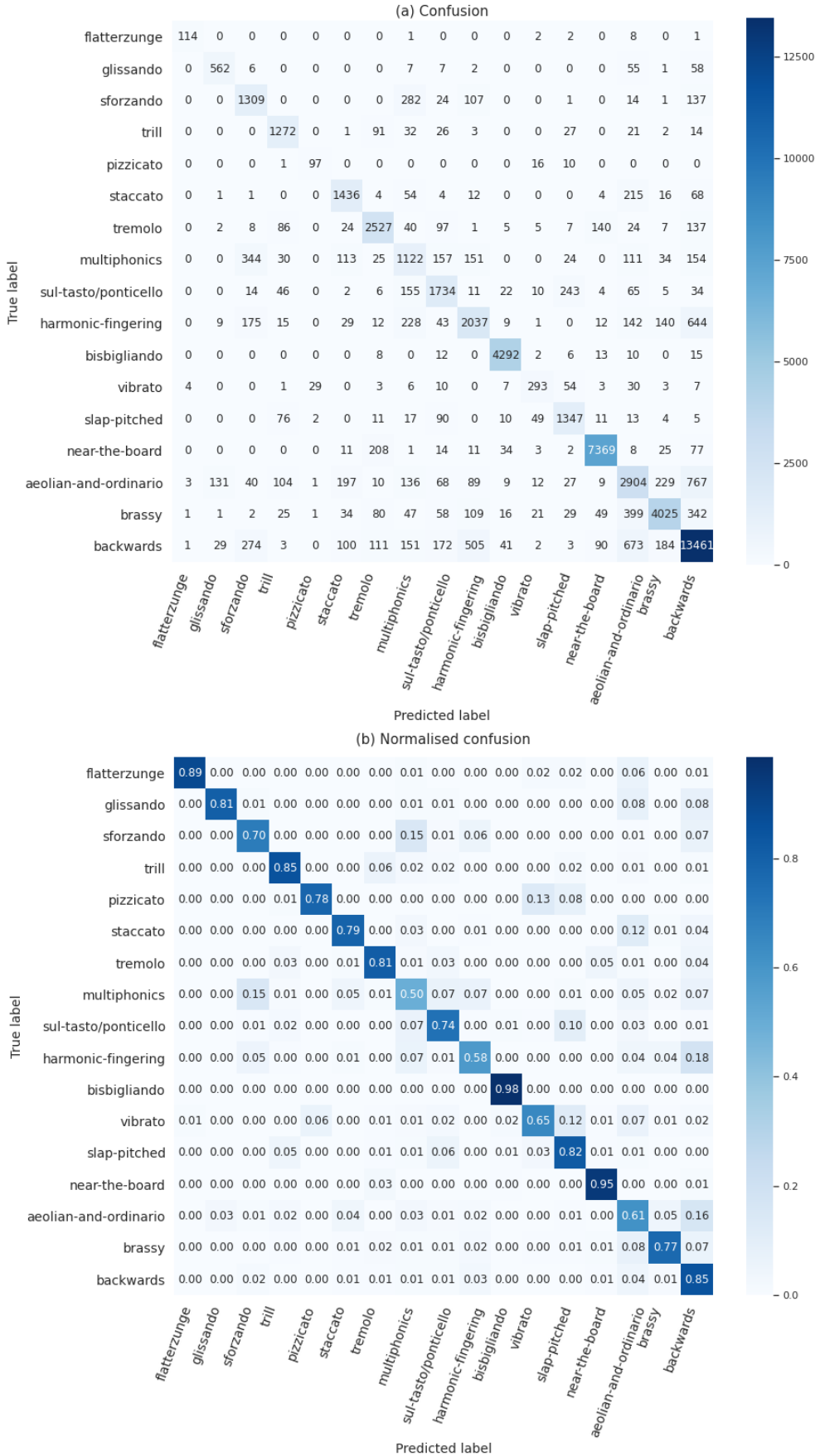

Fig. 3. Confusion matrices obtained for multi-class classification of techniques in the SOL dataset. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

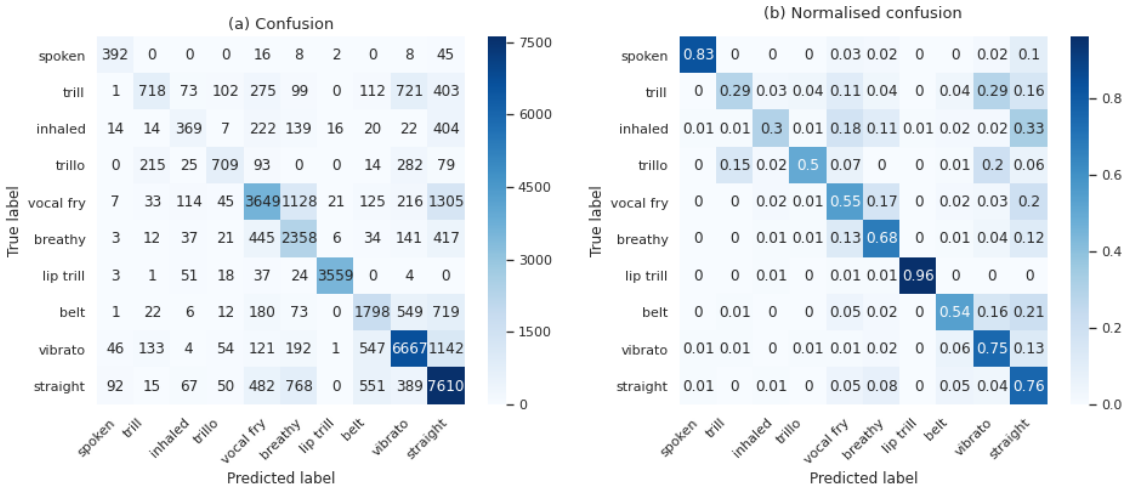

Fig. 4. Confusion matrices obtained for multi-class classification of ten vocal techniques. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

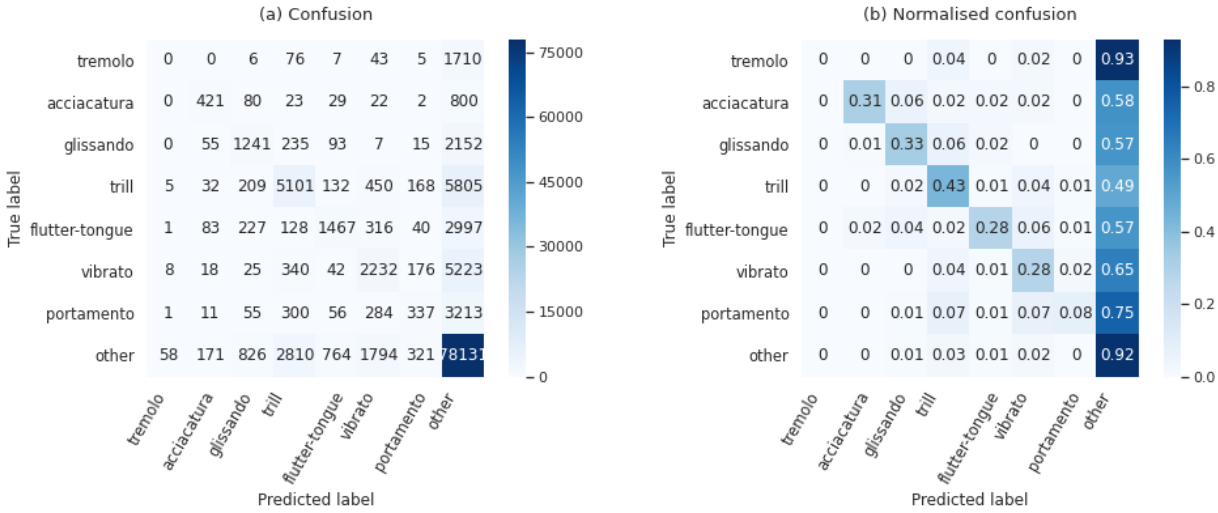

Fig. 5. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MFCC features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

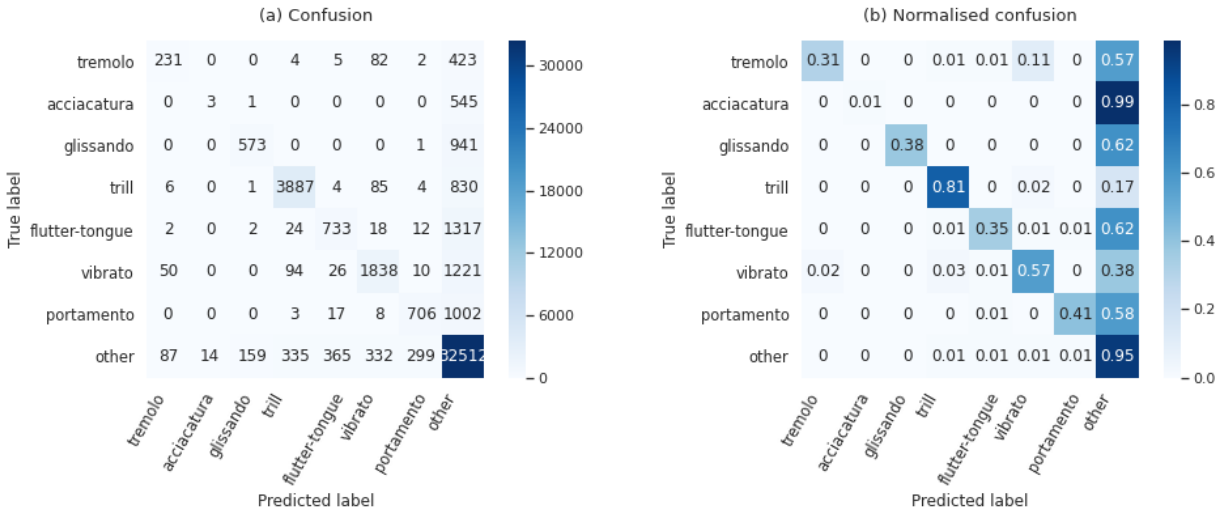

Fig. 6. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MPS features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.